MSc Mathematical and Computational Finance

Statistics and Financial Data Analysis - Problem Sheet 3

Data Analysis task: there is no need to display Python code in this exercise, but you need to display graphs (correctly labelled) and explain your conclusions, writing a coherent report on your model assumptions and diagnostics. Please restrict answers to Questions to 2 pages each.

1. Using `pandas_datareader`, download daily prices for symbols **GS**, **SPY**, **DIA**, **WMT**, **EURUSD=X** and **GBPUSD=X** from yahoo finance, using the `'Adj Close'` price for your analysis. In particular, use the period `'2018-01-01'-'2018-12-31'`.

   (a) Regress the daily log returns of **GS** on the daily log returns of all other predictors using a simple OLS model (no need to add or remove predictors for this exercise).

   (b) Use a cross-validation method in order to assess the MSE of the model for this OLS, using a 5-fold split to split data into training set and test set.

   (c) Next run a series of cross-validations (using 5-fold split) in order to choose the best lambda hyperparameter for a LASSO regression. Use the MSE of the test data for each lambda as your criteria. **Hints:** a) remember to set the model to be normalized and b) try lambda parameters using `lam = np.logspace(-8,-4,20)`.

   (d) Similarly, repeat the cross-validation exercise to select the best hyperparameter lambda for a Ridge regression. **Hints:** a) remember to set the model to be normalized and b) try lambda parameters using `lam = np.linspace(0.0005,0.05,20)`.

   (e) Using the results from c) and d), what is the test MSE for the optimal LASSO model? What about for the optimal Ridge model?

   (f) What are the coefficients of the optimal LASSO model? Has any of the parameters shrunk to zero?

   Note: when splitting the data for this exercise ignore any potential time series structure.

2. Consider the growth in manufacturing productivity in the UK, an index of which has been downloaded from the following site and saved under the CSV file name `'mfp.csv'`:

   http://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/labourproductivity/timeseries/a4ym/prdy

Using the quarterly data, investigate the possible autoregressive structure of the **growth** in productivity. Using the data provided, what predictions for the increase in productivity does it estimate for 2019-20? (use data up to Q2 2018 as in the CSV file). How has productivity been changing over the long-term?

By re-fitting your model over the periods 1997–2007 and 2008–2018, discuss what apparent changes have been seen in productivity growth following the 2008 financial crisis. Are these changes significant (both economically and statistically)? How does this change your predictions of productivity growth?

N.B. The following version of the t-test may prove useful. Suppose for populations $i = 1, 2$ you have independent estimates $\hat{x}_i$ of a parameter μ_i (e.g. the mean in an ARMA model) along with a squared standard error S_i^2 (i.e. standard errors of the estimator $\hat{x}_i$, not the standard deviation of the population), sampled with degrees of freedom d_i respectively. Recall that the appropriate degrees of freedom for an estimate in a regression is $n - k$, where k is the number of parameters fitted before calculating the variance. For an ARMA model, the estimated coefficients can be found with fit.params and their covariance matrix with fit.cov_params(), which has the squared standard errors on the diagonal (or by looking at the summary when fitted using ARMA in the statsmodels package). Then to test whether the parameter is the same in each model (without assuming they have the same standard errors) we can use the test statistic:

$$t = \frac{\hat{x}_1 - \hat{x}_2}{\sqrt{s_1^2 + s_2^2}}$$

Under the assumption that $\mu_1 = \mu_2$, t has an approximate t-distribution, with degrees of freedom ν given by the Welch–Satterthwaite approximation:

$$\nu = \frac{(s_1^2 + s_2^2)^2}{s_1^4/d_1 + s_2^4/d_2}$$

3. Using `pandas_datareader`, get daily volume traded data from **GOOG** for the period `'2012-01-04'- '2019-11-04'`. Verify for the presence of unit roots and proceed to address them. Testing for stationarity, choose the appropriate ARMA model, providing appropriate model diagnostics. **Extra for experts:** For your chosen model provide forecasts for volumes traded over the coming week, including prediction limits.